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Synthesis and characterization of ZnO-graphite binary nanocomposite for supercapacitor and photocatalyst application

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Abstract Zinc oxide (ZnO) is a wide-bandgap semiconductor that has been extensively studied for its applications in optoelectronic devices and energy storage systems, owing to its unique combination of electrochemical properties, high surface area, and structural stability. ZnO exhibits excellent catalytic and charge transport properties, synergistically interacting with graphite, a conductive carbon material renowned for its superior electron mobility and chemical stability. This composite leverages the strengths of both components, enhancing performance in applications such as supercapacitors, lithium-ion batteries, and photocatalysis. The formation of crystalline ZnO nanoparticles with a flower-shaped morphology, consisting of clusters of nanopetals, has been confirmed through X-ray diffraction (XRD), and scanning electron microscopy (SEM) analyses. Symmetric supercapacitors were fabricated using ZnO and its composite with graphite, and their electrochemical performance was evaluated using a 6 M KOH electrolyte. Cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy demonstrated that ZnO/Graphite composite electrodes exhibited superior charge storage capabilities compared to ZnO electrodes. The ZnO/graphite-based electrodes achieved a specific capacitance of 136 Fg^{-1} at a scan rate of 10 mVs^{-1} and the energy density of 20.8 $Whkg^{-1}$ at a power density of 797.64 Wkg^{-1}. The porous structure and high surface area of the ZnO nano petals, combined with the complementary properties of graphite, create a synergistic effect that enhances the charge storage mechanism, resulting in improved specific capacitance and overall performance. Additionally, ZnO demonstrated good photocatalytic activity under UV–vis light irradiation, while the ZnO/Graphite composite exhibited significantly enhanced photocatalytic activity under visible light irradiation. This enhancement is attributed to the synergistic interaction between ZnO and the graphite-like carbon layer. A possible mechanism for the improved photoactivity under visible light was also discussed to provide further insights into the material's behavior.

1 Introduction

As fossil fuel consumption continues to increase rapidly, the need for efficient energy conversion and storage has become a pressing issue for current and future research. Globally, approximately 36.5 billion barrels of oil, 3.9 trillion cubic meters of natural gas, and 7.8 billion tons of coal are consumed annually [1–4]. This heavy reliance on fossil fuels leads to severe consequences, including energy shortages, greenhouse gas emissions, air pollution, environmental degradation, and water contamination, all of which endanger biodiversity and disrupt ecosystems [5, 6]. To meet the world's growing energy demands, the development of cost-effective materials for energy conversion and storage is crucial for large-scale industrial use. Transitioning to sustainable energy sources such as solar, wind, hydroelectric, and geothermal power, along with implementing energy-efficient technologies, is essential to mitigating these challenges [7–9]. This has led to a growing emphasis on utilizing nanomaterials and their binary composites, which exhibit exceptional properties that make them highly suitable for various applications, including supercapacitors, sensors, electrochemical hydrogen generation, batteries, and photocatalysts [10–13]. Electrochemical supercapacitors (SCs) have emerged as a novel energy storage technology, widely utilized in electric vehicles, aerospace applications, power sources, portable electronics, and other devices. They have garnered significant interest due to their high power density (10 kW kg^{-1}), simple structure, extended lifespan ($> 10^5$ cycles) compared to batteries, environmental friendliness, rapid charge–discharge capability (within seconds), and greater energy density than conventional dielectric capacitors. However, a key challenge with supercapacitors is their relatively low-energy density (5–20 W h kg^{-1}), which is approximately 20–40 times lower than that of lithium-ion batteries (100–265 W h kg^{-1}), limiting their potential to meet future energy demands. To overcome this limitation, extensive research efforts have been directed toward enhancing energy density by increasing specific capacitance and expanding voltage windows. Supercapacitors are classified into two main types based on their charging mechanisms: electrochemical double-layer capacitors (EDLCs) and pseudocapacitors. In EDLCs, charge storage

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occurs through reversible ion adsorption at the interface between the electrode and electrolyte. In contrast, pseudocapacitors store charge through Faradaic redox reactions occurring on the electrode surface [14–16].

Metal oxides such as zinc, iron, cobalt, and manganese exhibit unique properties due to their nanoscale size, high surface area, and quantum effects. Their hybrid forms have been extensively studied for applications in supercapacitors and photocatalysis. Among them, nanostructured zinc oxide (ZnO) can be synthesized in various morphologies, including nanorods, nanowires, nanobelts, and nanohelices. ZnO possesses remarkable electrical, optical, and electrochemical properties due to its wide band gap (3.37 eV) and high exciton binding energy (60 meV).

However, its rapid charge carrier recombination, photo corrosion, and limited absorption of visible light hinder its widespread use. To address these limitations, extensive research has focused on improving ZnO's properties through doping and co-doping with metal, non-metal, and carbonaceous materials such as graphite, graphene, and black carbon materials. ZnO-based composites have also shown promise in energy storage and conversion applications due to their enhanced electrochemical activity, cost-effectiveness, and environmentally friendly nature [17].

Eswaramoorthy et al. examine the electrochemical performance of ZnO/MWCNTs nanocomposite material in a 1 M Na_2SO_4 electrolyte. In three three-electrode configuration, the CV and GDC curves exhibit good reversibility and attain a specific capacitance of 115.7 F/g at a scan rate of 5 mV/s. Selvakumar et al. [18] evaluated the electrochemical properties of the ZnO-AC (1:1) nanocomposite electrodes, and the supercapacitor performance has been studied using cyclic voltammetry (CV) in 0.1 M Na_2SO_4 electrolyte solution. The nanocomposite electrode showed a specific capacitance of 123 F/g at a scan rate of 5 mV/s [19].

On the other hand, incorporating graphite into ZnO nanoparticles has been found to improve conductivity and stability. Notably, graphite, being carbon-based, increases surface area, provides more active sites, and facilitates efficient photoinduced charge transfer compared to bulk ZnO, making it a highly promising material for photocatalytic and supercapacitor applications [20]. Kuruvilla et al. synthesized 3D Ni-Co/ZnO incorporated graphite (G-NC/ZnO) nanocomposite using the hydrothermal method. The nanocomposite achieved a maximum photocatalytic degradation efficiency of 96.6% within 150 min of visible light irradiation at 475 nm and also exhibited high stability and recyclability [21].

In this study, ZnO/Graphite composites were successfully synthesized using a combination of sol–gel and physical grinding techniques. The synergistic properties of the ZnO/Graphite composite resulted in the development of a hybrid electrode material with an increased number of active sites and distinct low-energy planes. This unique structure enhances electron–ion transport, thereby improving both energy storage capacity and photocatalytic performance. The physical grinding method plays a crucial role in expanding the electroactive surface area of the electrode, reducing the electron and ion travel distance during charge–discharge cycles, and ultimately enhancing the overall efficiency of the device. Additionally, the ZnO/Graphite composite is expected to improve electrode kinetics, leading to better performance and long-term cycling stability. To fabricate the composite, ZnO nanoparticles were first synthesized using the sol–gel method. Subsequently, the ZnO/Graphite composite was prepared through a one-step physical grinding technique. The electrochemical performance of the materials was evaluated using a two-electrode configuration, where the composite exhibited a specific capacitance of 136 F/g at 10 mV/s. Furthermore, the composite demonstrated impressive energy and power densities of 20.8 $Whkg^{-1}$ at 797.64 Wkg^{-1}. The photocatalytic activity of the ZnO/Graphite composite was also assessed, revealing that it effectively degraded 86.2% of the absorption peaks of the methylene blue (MB) dye. These findings highlight the ZnO/Graphite composite as a promising material for both energy storage and photocatalytic applications.

2 Experimental method

2.1 Materials preparation

All of the reagents were of analytical purity, purchased from Sigma, and did not require additional purification before use. We used our high-purity deionized water as a solvent.

2.2 Synthesis of ZnO

The ZnO was synthesized using zinc acetate dihydrate ($Zn(CH_3COO)_2{\cdot}2H_2O$, 99.99%), sodium hydroxide pellets (NaOH, 98%), and distilled water. We use sol sol–gel method to synthesize the ZnO nanoparticles. Firstly, zinc acetate dihydrate and sodium hydroxide pellets were added separately in two different beakers with 40 ml of distilled water (DI). To prepare ZnO nanoparticles, the sodium hydroxide (NaOH) solution was slowly added dropwise to the first solution containing 0.5 M zinc acetate solution under constant magnetic stirring. The resulting reaction mixture was kept at 60 °C for 3 h under continuous stirring to get a white precipitate gel. Subsequently, the precipitate was washed several times and dried at 80 °C, which yields a fine white powder. Finally, the sample was annealed at 800 °C to obtain ZnO nanoparticles.

2.3 Synthesis of ZnO/graphite

The ZnO/Graphite composite was synthesized using a simple and cost-effective physical blending method. In this procedure, pre-synthesized ZnO nanoparticles and graphite powder were taken in a 1:1 weight ratio. The materials were placed in an agate pestle and mortar and thoroughly ground for approximately 30–45 min to ensure uniform mixing and intimate contact between ZnO particles and graphite layers. Continuous grinding facilitated the even distribution of ZnO nanoparticles on the surface of graphite sheets, forming a homogeneous composite mixture. The obtained mixture was then dried at 80 °C for 6 h in a hot air oven to remove any residual moisture. After drying, the composite powder was stored in airtight glass vials to prevent contamination and moisture absorption. The resulting ZnO/Graphite was then characterized by different methods.

2.4 Materials characterization

The structural analysis of the samples was performed using an X-ray diffractometer equipped with Cu Kα radiation, with a wavelength of $\lambda = 1.5418$ Å. The particle size was also calculated from the XRD data. Additionally, the morphology was analyzed using SEM (Bruker). Transmittance and absorbance measurements were carried out with a UV–vis spectrophotometer (JASCO, model: V-770PC) with wavelengths ranging from 200 to 1000 nm. The quantitative optical analysis calculates the energy band gap (E_g) using a Tauc plot. The photocatalytic activity of ZnO/Graphite composites was tested for the degradation of methylene blue (MB), a non-biodegradable organic dye pollutant, using a 450 W high-pressure mercury lamp as the irradiation source.

The photocatalytic performance of ZnO/Graphite composites, MB (methylene blue) dye, a persistent and non-biodegradable organic pollutant, was selected, and its degradation efficiency was tested under visible light irradiation at 420 nm [31]. A 20-ppm solution of MB dye was prepared, and 0.2 g of ZnO/Graphite composites (photocatalysts) was introduced into the solution. The photocatalyst mixture was immersed in an aqueous solution and stirred continuously under dark conditions for 35 min to establish an adsorption–desorption equilibrium. The sample suspension was then exposed to visible light using a 450 W high-pressure mercury lamp, and 10 mL of the sample was collected from a customized quartz reactor at regular intervals. The collected samples were centrifuged at 6000 rpm for 5 min and analyzed using a UV–vis spectrophotometer (JASCO, model: V-770PC). To evaluate the stability of the organic pollutant MB dye, degradation analysis was conducted without adding a catalyst, maintaining a constant pH level.

2.5 Electrochemical measurements

During the fabrication process, the electrodes were prepared using a blend of active electrode material, activated carbon, and PVDF in a weight ratio of 70:20:10. This mixture was then combined with *N*-methyl-2-pyrrolidone solvent and continuously ground until a homogeneous slurry was formed. The resulting slurry was applied onto nickel foam and dried at 85 °C for 12 h to ensure proper adhesion and stability. For the electrolyte preparation, a Whatman filter paper was soaked in a 6 M KOH solution and then left to dry at room temperature for 1 h. A symmetric supercapacitor was assembled by sandwiching the dried Whatman filter paper between two identical electrodes under hydraulic pressure. The nickel foam used in the electrodes had an area of 1.2 cm^2. The supercapacitive performance of the fabricated device was evaluated using a CHI760E electrochemical workstation. Additionally, the catalytic activity of the ZnO/Graphite composite was thoroughly examined by monitoring the photodegradation of an aqueous methyl orange (MO) solution under both natural sunlight and UV irradiation. This dual-exposure method allowed for the assessment of the ZnO/Graphite composite photocatalytic efficiency in real-world conditions as well as under controlled UV light. The degradation rate was tracked over time, demonstrating the potential of ZnO/Graphite composite as an effective photocatalyst for environmental remediation applications.

3 Results and discussions

3.1 Structural analysis:

The X-ray diffraction (XRD) patterns of ZnO and ZnO/Graphite composite materials are presented in Fig. 1. Figure 1a displays the diffraction pattern of ZnO particles, showing prominent peaks at 2θ positions of 31.97°, 34.61°, 36.48°, 47.74°, 56.68°, 63.14°, 66.61°, 68.05°, 69.31°, 72.68°, and 77.12°. These peaks correspond to the (100), (002), (101), (102), (110), (103), (112), (201), (004), and (202) crystal planes, respectively, confirming the hexagonal wurtzite structure of the synthesized ZnO particles (JCPDS No. 36-1451).

Figure 1b presents the XRD pattern of the ZnO/Graphite composite. Notably, two distinct peaks appear at 26.76° and 54.96°, corresponding to the (002) and (004) crystal planes of graphite, respectively. The presence of characteristic peaks from both ZnO and graphite in the XRD pattern confirms the successful formation of a binary-phase ZnO/Graphite composite.

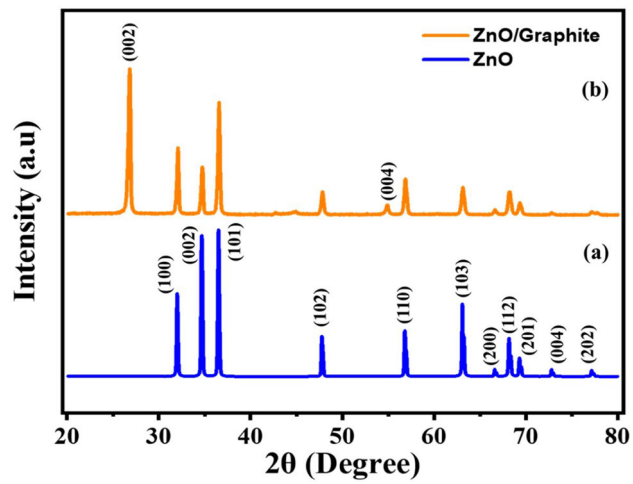


Fig. 1 **a** Shows the XRD pattern of ZnO, **b** ZnO/Graphite

3.2 Morphological analysis

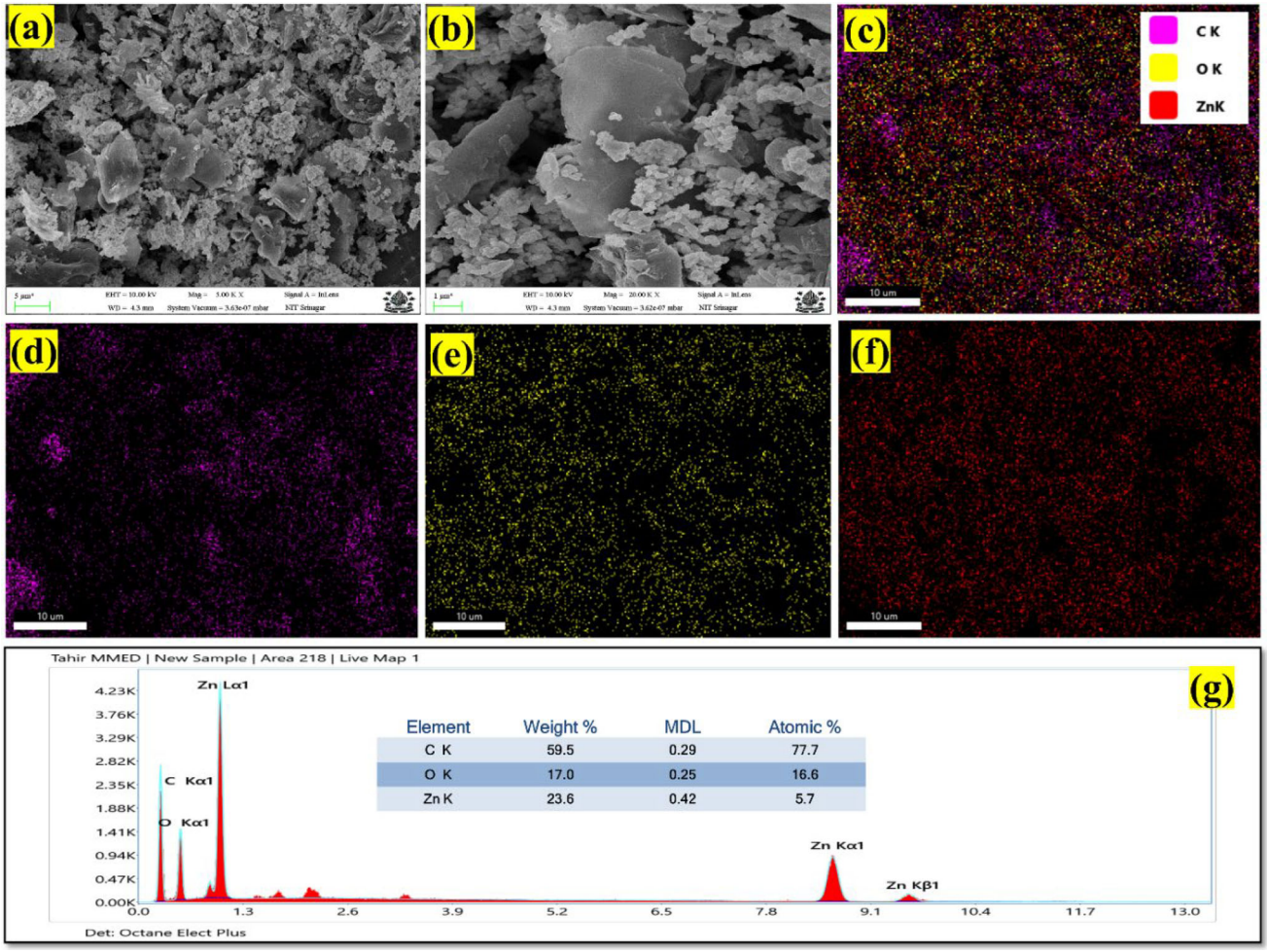
The surface morphology of the synthesized ZnO/Graphite composite was analyzed using scanning electron microscopy (SEM), as shown in Fig. 2. Figure 2a and b depict the composite's low- and high-resolution morphology, respectively. The SEM images reveal that the matrix consists of micro-sized graphite particles with a layered structure-like morphology, while ZnO particles are uniformly distributed on the graphite surface. The incorporation of ZnO particles into the graphite matrix enhances the composite's surface area and porosity. This structural feature is crucial as it contributes to improved electrochemical performance and photocatalytic efficiency by providing more active sites for charge transfer and reaction processes. Figure 2c illustrates the elemental mapping of a selected region of the ZnO/Graphite composite, confirming the homogeneous distribution of carbon (C), oxygen (O), and zinc (Zn) elements throughout the composite matrix. This mapping provides strong evidence for the uniform integration of ZnO nanoparticles with the graphite surface. The corresponding elemental maps shown in Fig. 2d–f further display the individual spatial distributions of carbon, oxygen, and zinc, respectively. The carbon map highlights the continuous graphite framework, while the oxygen and zinc maps confirm the presence and even dispersion of ZnO nanoparticles over the graphite substrate. Furthermore, Fig. 2g presents the energy-dispersive X-ray spectroscopy (EDS) analysis of the ZnO/Graphite composite. The EDS spectrum reveals prominent peaks corresponding exclusively to carbon, oxygen, and zinc, thereby validating the successful synthesis of the ZnO/Graphite composite without any noticeable impurity peaks. The absence of additional elements confirms the high purity of the prepared material. This compositional analysis substantiates that ZnO nanoparticles were effectively anchored onto the graphite sheets, ensuring a uniform and stable composite structure suitable for further electrochemical applications.

3.3 Electrochemical analysis

The electrochemical performance of the synthesized electrode materials was evaluated in a 1 M sodium sulfate electrolyte, as illustrated in Fig. 3. Figure 3a presents the cyclic voltammetry (CV) curves of the ZnO/Graphite composite at varying scan rates (10–100 mV/s) within a potential range of − 0.6 to 0.6 V. All CV curves exhibit distinct redox behavior, with the current density increasing as the scan rate rises, indicating a pseudocapacitive response. At lower scan rates, redox peaks are observed around 0.17 V in the positive current density region, representing the oxidation process, while cathodic peaks appear near − 0.16 V in the negative current density region, corresponding to the reduction process. The specific capacitance values were determined from the CV curves using the appropriate mathematical expression (1).

$$C_{SP} = \frac{1}{mV(Va - Vc)} \int_{0.6}^{-0.6} I(V)dV \tag{1}$$

The integration term represents the area enclosed by the CV curve, while m denotes the mass of the active material loaded onto the electrode, and V refers to the scan rate. Va and Vc correspond to the anodic and cathodic potentials, respectively, while I represents the instantaneous current. As shown in Fig. 3b, the specific capacitance (C_{SP}) of the ZnO/Graphite composite was found to decrease with increasing scan rate. This decline in C_{SP} at higher scan rates is likely due to the limited time available for complete redox reactions to take place. The highest recorded C_{SP} value was 136 F/g at a scan rate of 10 mV/s; the comparison of specific capacitance with the previous study is shown in Table 1. Additionally, as the scan rate increased, the anodic peak shifted toward higher potentials, while the cathodic peak moved toward lower potentials, leading to a broader potential difference. This behavior is characteristic of quasi-reversible kinetic processes. To further investigate the electrochemical performance of the electrode, the power law was applied to analyze the charge storage mechanism. This mechanism consists of three primary components: (1) faradaic surface redox reactions, (2) intercalation pathways, both contributing to pseudocapacitive behavior, and (3) interfacial charge storage associated



Element	Weight %	MDL	Atomic %
C K	59.5	0.29	77.7
O K	17.0	0.25	16.6
Zn K	23.6	0.42	5.7

Fig. 2 **a, b** SEM images of ZnO/Graphite low and high resolution, **c** overall mapping of composite, **d-f** display the mapping of carbon, oxygen, and Zinc, **g** shows the elemental composition of binary composite

with the formation of an electric double layer (EDL). The power law equation, used to assess these charge storage processes, is expressed as Eq. (2) [14, 22]:

$$i = aV^b \tag{2}$$

where a and b are adjustable parameters, v is the scan rate in mV/s, and i represents the peak current. The b value is determined from the slope of the log (i) vs. log (v) plot, providing insight into the dominant charge storage mechanism. Reports suggest that a magnitude of approximately 0.5 for b is linked to an intercalation mechanism, while a value approaching 1 indicates capacitive behavior. In the case of ZnO graphite, it has been observed that the value of ~0.66, as shown in Fig. 3c, which, as per the report attributed to the dominant intercalation mechanism.

Moreover, the contribution ratio of the diffusive and capacitive charge storage mechanisms can be estimated using Dunn's method, as expressed in Eq. (3).

$$i = k_1 v + k_2 v^{1/2} \tag{3}$$

where i represents the current at a fixed scan rate (v), $k_1 v$ and $k_2 v^{1/2}$ correspond to the current contributions from the capacitive and diffusive (intercalation-controlled) processes, respectively. At a low scan rate of 10 mVs^{-1}, the binary composite exhibits a capacitive contribution of approximately 13%, indicating that the charge storage process is predominantly governed by the diffusive mechanism. This dominance of diffusion-controlled behavior is clearly depicted in the bar graph Fig. 3d. As the scan rate increases, the proportion of the diffusive contribution gradually decreases because the ions have insufficient time to diffuse and intercalate into the bulk of the electrode material. Instead, the charge storage process becomes increasingly dominated by surface-controlled capacitive reactions. This trend is consistent with the broadening and shifting of the redox peaks observed in the CV profiles at higher scan rates. Furthermore, the capacitive and diffusive contributions at a representative scan rate of 40 mV s^{-1} are illustrated

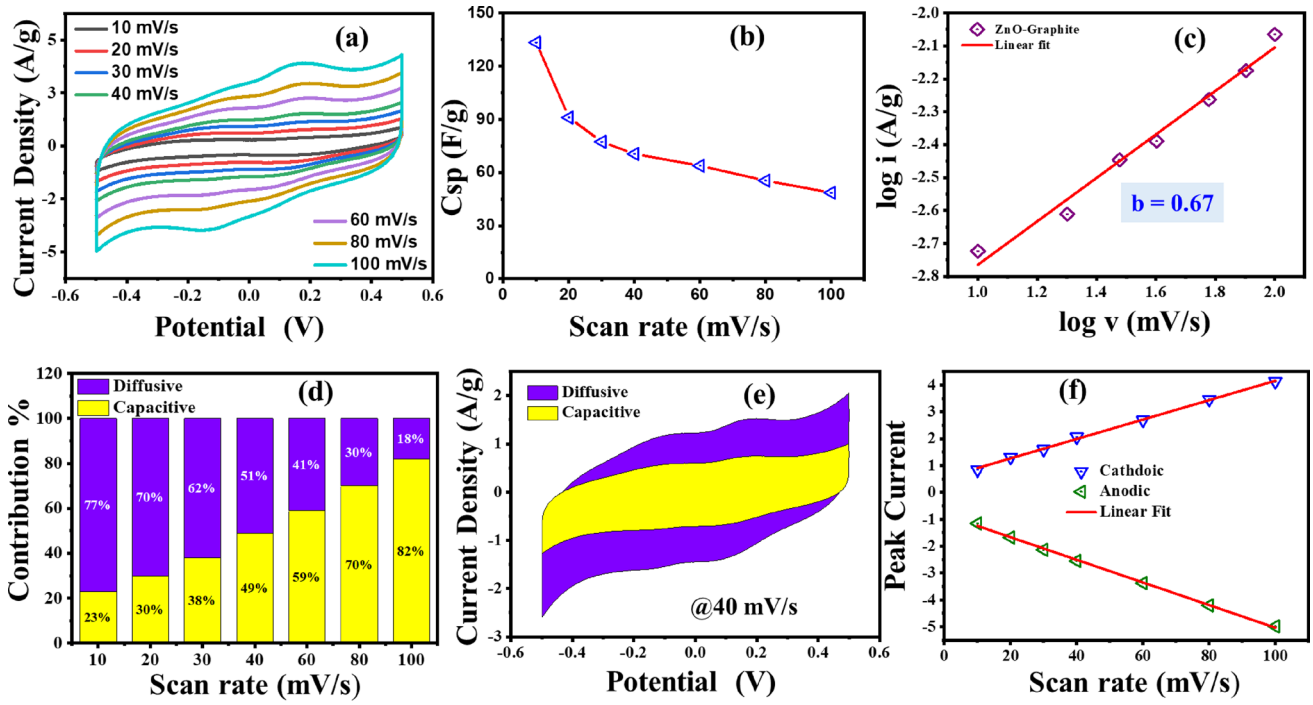


Fig. 3 **a** CV curves of ZnO Graphite at different scan rates, **b** sepecific capacitance versus scan rates, **c** Log(i) versus Log(v), **d** percentage contribution of capaitive and diffusive at different scan rate, **d** capacitive and diffusive contribution at fix scan rate, and **e** peak current contribution form cathodic and anodic

Table 1 Contrast between the as-synthesized ZnO graphite-based electrode and earlier reports

Electrode	Electrolytes	Specific capacitance (F/g)	Scan rate (mV/s)	References
ZnO NPs	3 M KOH	86	2	[26]
ZnO/AC	1 M Na_2SO_4	68.2	2	[27]
ZnO/FMCN	0.1 M TBAPC/DMF	53	5	[28]
ZnO nanorod	0.5 M Na_2SO_4	36	5	[29]
ZNO/RGO	0.1 M Na_2SO_4	95.0 F/g,	10	[30]
AC–ZnO (1:1)	0.1 M Na_2SO_4	105	10	[19]
ZnO/MWCNTs	1 M Na_2SO_4	115.7 F g^{-1}	5	[18]
ZnO graphite	6 M KOH	136 F/g	10	Present work

in Fig. 3e, highlighting the coexistence of both mechanisms. Figure 3f presents the linear relationship between the peak current and the square root of the scan rate, further confirming the mixed charge storage behavior of the binary composite electrode.

Additionally, the electrochemical properties of the nanocomposite were evaluated through galvanostatic charge–discharge (GCD) measurements at current densities ranging from 2 to 6 A/g, as shown in Fig. 4a. It can be observed that all the charge–discharge curves are symmetric. The charge–discharge time decreases as the applied current density increases. The specific capacitance (*Csp*) was calculated from the charge–discharge data using the following Eq. (4) [23]:

$$C_{SP} = \frac{I\Delta t}{m\Delta V} \tag{4}$$

where I is the discharge current, Δt is the discharge time in seconds, m is the mass of the prepared electrode materials, and ΔV is the recorded potential window. The specific capacitance estimated from the GCD profile is approximately 100 F/g at a current density of 2 A/g. The electrochemical performance of the ZnO/Graphite composite was further assessed using electrochemical impedance spectroscopy (EIS). To validate the experimental EIS results, the Nyquist plot was simulated with an equivalent electrical circuit (as shown in the inset) using ZSimpWin (version 2.0). The Nyquist plot displays a semi-circular shape at high frequencies, indicating the absence of an inductive coupling channel in ZnO (no negative quadrature). However, the slight increase in the circular arc from the origin suggests resistive coupling in this region. The bulk resistance (*Rb*) and charge transfer resistance (*Rct*) were extracted by observing the intercepts on the x-axis at both high- and low-frequency regions of the Nyquist plot. *Rb* represents the resistance due to the interfacial and intrinsic components, while *Rct* is associated with the impedance from the double-layer capacitance and Faradic components (Fig. 5).

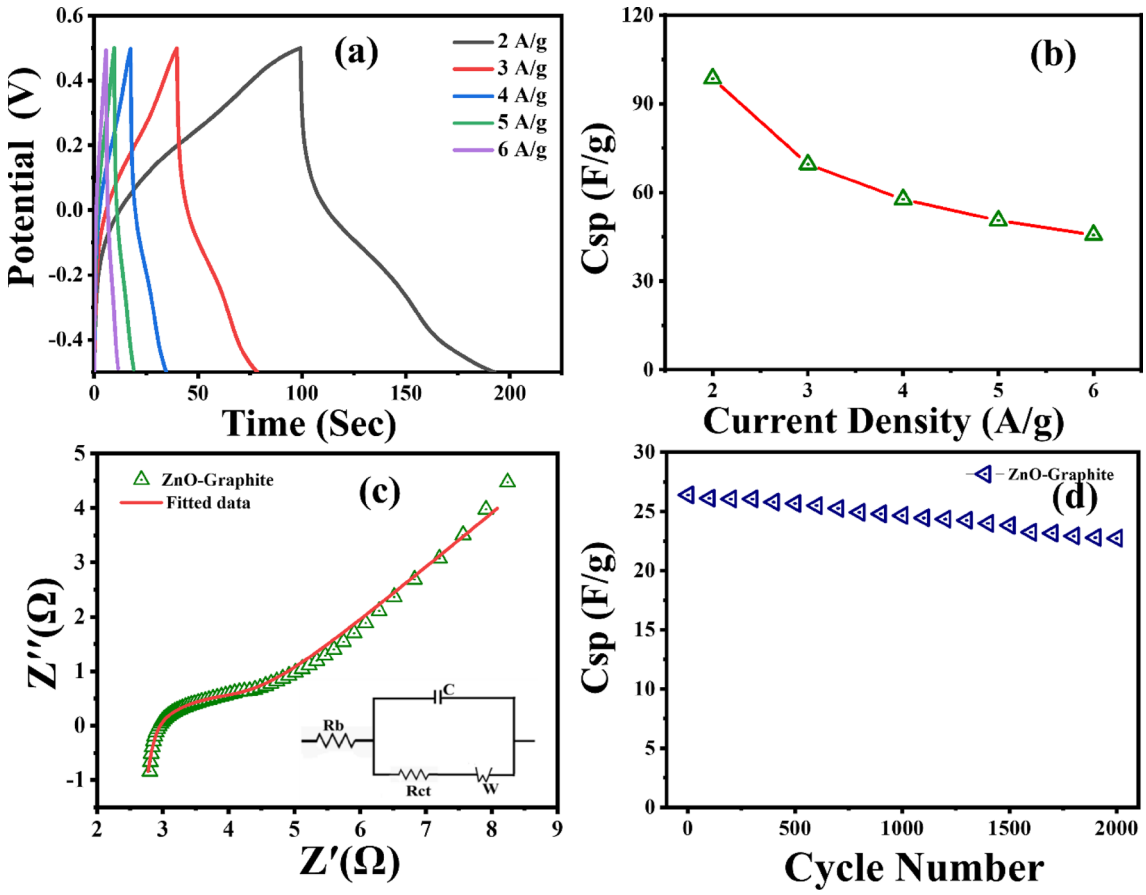


Fig. 4 **a** GCD plot of ZnO-Graphite at different current densities, **b** specific capacitance vs current density, **c** impedence spectra of prepared electrode materials, and **d** cyclic stability

The performance of the supercapacitor device was evaluated by making a symmetric supercapacitor assembly using the ZnO/Graphite composite as positive and negative electrode material, as illustrated in Fig. 6a. Meanwhile, the energy and power density are the key parameters for the supercapacitor's performance, as illustrated in the Ragon plot in Fig. 6b, calculated using relations 5 and 6 [24, 25].

$$Ed = \frac{C_{sp} * V^2}{7.2} \tag{5}$$

$$Pd = \frac{Ed}{t} * 3600 \tag{6}$$

The ZnO/Graphite composite delivers an energy density of 20.8 Wh/kg with a power density of 797.64 W/kg at a current density of 2 A/g. This enhancement in the energy density of the composite might be attributed to the synergistic effect between the ZnO and graphite particles, which increases the conductivity and stability and accelerates ions/electrons moments. Also, the large number of active sites, rapid electron transfer, and large potential window. Nickel foam was used with conducting outlets for connection to prepare a setup for the demonstration of practical applicability. The two cells arranged in a series combination are connected to the positive and negative terminals of the LED, as shown in Fig. 6b. The LED glowed for more than 3 min. This demonstration proved the concept of implementing ZnO/Graphite as a realistic super.

3.4 Photocatalytic experiment

To evaluate the photocatalytic performance of ZnO/Graphite composites, MB (methylene blue) dye, a persistent and non-biodegradable organic pollutant, was selected, and its degradation efficiency was tested under visible light irradiation at 420 nm [31]. A 20-ppm solution of MB dye was prepared, and 0.2 g of ZnO/Graphite composites (photocatalysts) was introduced into the solution. The photocatalyst mixture was immersed in an aqueous solution and stirred continuously under dark conditions for 35 min to establish an adsorption–desorption equilibrium. The sample suspension was then exposed to visible light using a 450 W high-pressure mercury lamp, and 10 mL of the sample was collected from a customized quartz reactor at regular intervals. The collected samples were centrifuged at 6000 rpm for 5 min and analyzed using a UV–vis spectrophotometer (JASCO, model: V-770PC). To

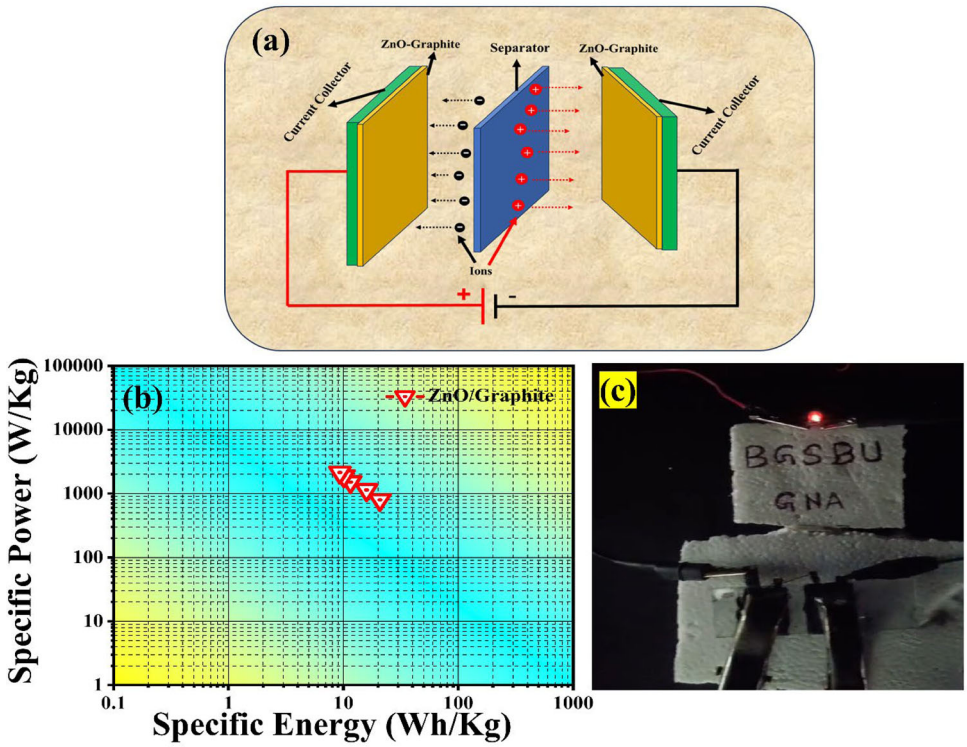


Fig. 5 **a** Schematics of working ZnO/Graphite symmetric supercapacitor, **b** specific energy and power density, and **c** photograph of demonstration of LEDs with two symmetric cells in series

evaluate the stability of the organic pollutant MB dye, degradation analysis was conducted without adding a catalyst, maintaining a constant pH level.

3.5 Photocatalytic activity

The photocatalytic performance of the synthesized ZnO/Graphite composites was assessed by evaluating their efficiency in degrading MB dye under visible light irradiation. The degradation efficiency of the nanocomposites was tested under identical conditions, varying only the irradiation time. Figure 6a illustrates the UV–visible spectra of MB recorded at different time intervals during the photocatalytic activity [32]. The photodegradation performance or efficiency (η) of the prepared ZnO/Graphite composites was determined using Eq. (7) [33].

$$\eta(\%) = \left(1 - \frac{C_t}{C_0}\right) \times 100 \tag{7}$$

C_t and C_0; where C_0 represents the initial concentration of MB and C_t is the concentration of MB after degradation, respectively. According to the obtained results, the photocatalysis reduced 86.2% of the absorption peaks of (MB) dye, showing that ZnO/Graphite composites have shown significant photocatalytic activity. The relationship between the photocatalytic reaction rate and MB dye concentration can be described using the Langmuir–Hinshelwood kinetic model [34], which is expressed by the following Eq. (8)

$$\ln \frac{A_0}{A_t} = t \times k \tag{8}$$

"k" represents the rate at which a photocatalyst deteriorates when exposed to light (Fig. 7).

Figure 6a and b illustrate the degradation rates and kinetics of MB dye by ZnO/Graphite composite, respectively, under visible light irradiations. The slope values in Fig. 6a and b can be used to compute the rate constant (k, linear regression analysis of the plot) for MB and ZnO/Graphite composite photodegradation using the pseudo-first-order kinetics reaction, from the calculated rate constant value, it was noticed that the photodegradation is 86.2% as compared with the other photocatalysts. When a photon strikes the mixture of ZnO/Graphite composite and MB dye, the electrons are stimulated from the valence band (VB) to the conduction band (CB), according to Eq. (4). This photon absorption leads to the production of electron–hole pairs. Excited electrons in the conduction band and produced holes in the valence band migrate to the ZnO/Graphite composite surface. Electrons participate in reduction reactions, which often involve oxygen molecules to make reactive oxygen species, whereas holes engage in oxidation reactions, which frequently involve water or hydroxide ions to produce hydroxyl radicals [5, 8, 35, 36]. This unique result is attributed to the presence of graphite nanoparticles that decreased the recombination of electron–hole pairs and to the effect of ZnO in graphite that increases the light-harvesting ability, which greatly promoted the photoactivity of the catalyst. The enhancement of production

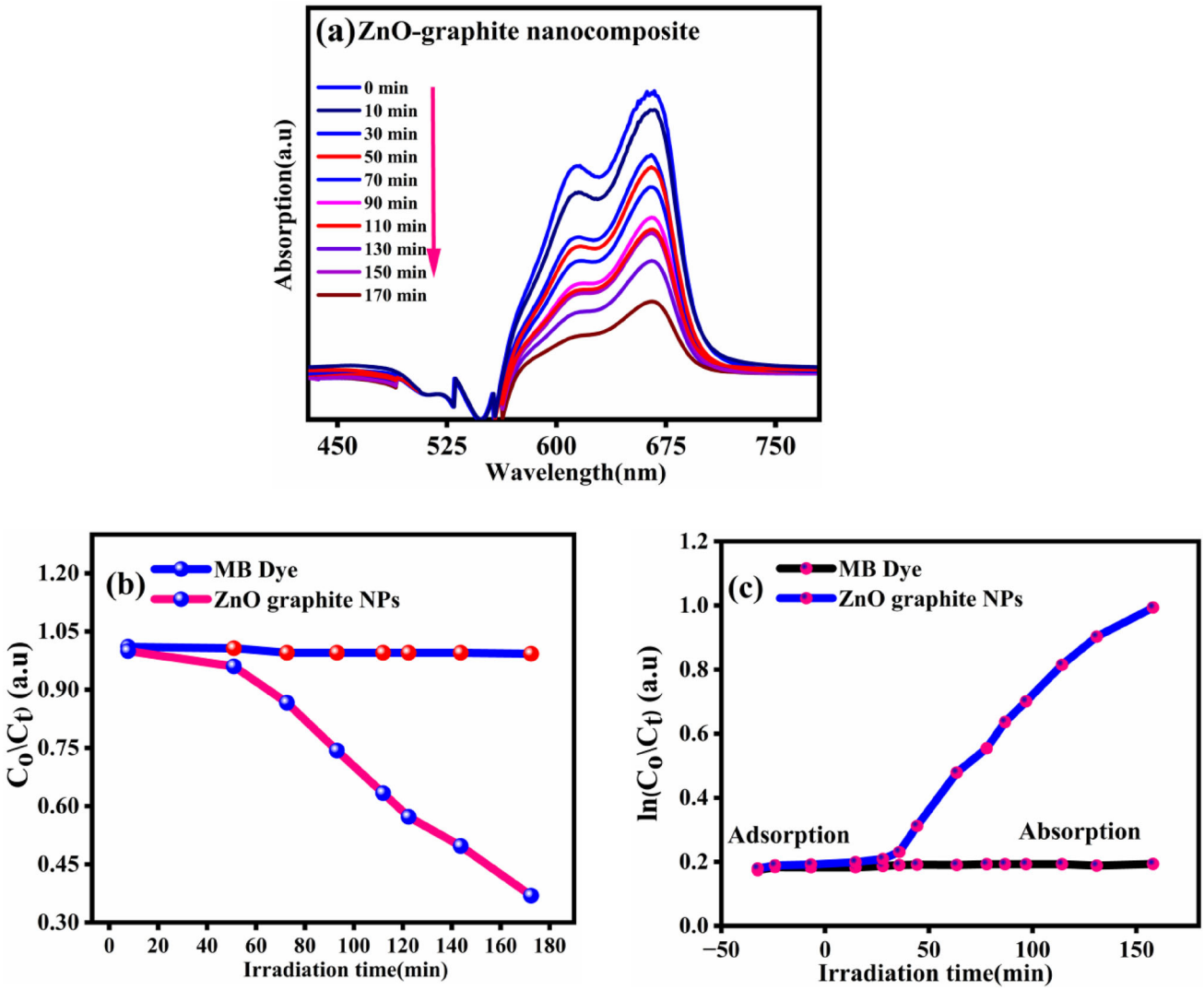


Fig. 6 **a** The UV-visiblespectra of MB recorded at different time intervals during the photocatalytic activity, **b** the degradation rates, and **c** kinetics of MB dye by ZnO/Graphite composite under visible light irradiations

hydroxyl radicals was influenced by the synergistic effect of the graphite on the activity of the semiconductor ZnO by electron and hole trapping [4, 15, 37, 38].

3.6 Mechanism

The photocatalytic mechanism of ZnO/graphite composite has been extensively studied by various researchers. The photocatalytic degradation of MB dye by ZnO-graphite composite under visible light irradiation can be explained through three proposed mechanisms which involves the following key steps: (i) absorption of photon energy with energies greater than the optical band gap of the semiconductor material; (ii) generation of electron–hole (e–h) pairs within the semiconductor upon photon absorption; and (iii) participation of the photogenerated (e–h) pairs in surface oxidation and reduction reactions, facilitating the degradation process on the photocatalyst surface [3, 35–37, 39, 40]. The semiconductor initially absorbs photons with energies exceeding its band gap, exciting electrons from the valence band to the conduction band and generating electron–hole pairs. The charge carriers then migrate to the surface of the semiconductor, where they facilitate reduction and oxidation reactions. Electrons in the conduction band reduce oxygen molecules to generate reactive oxygen species, while holes in the valence band oxidize water or hydroxide ions, forming highly reactive hydroxyl radicals. These reactive species then interact with contaminants, breaking them down into less harmful chemicals and achieving the desired photocatalytic degradation. The photocatalytic degradation of MB by ZnO/Graphite

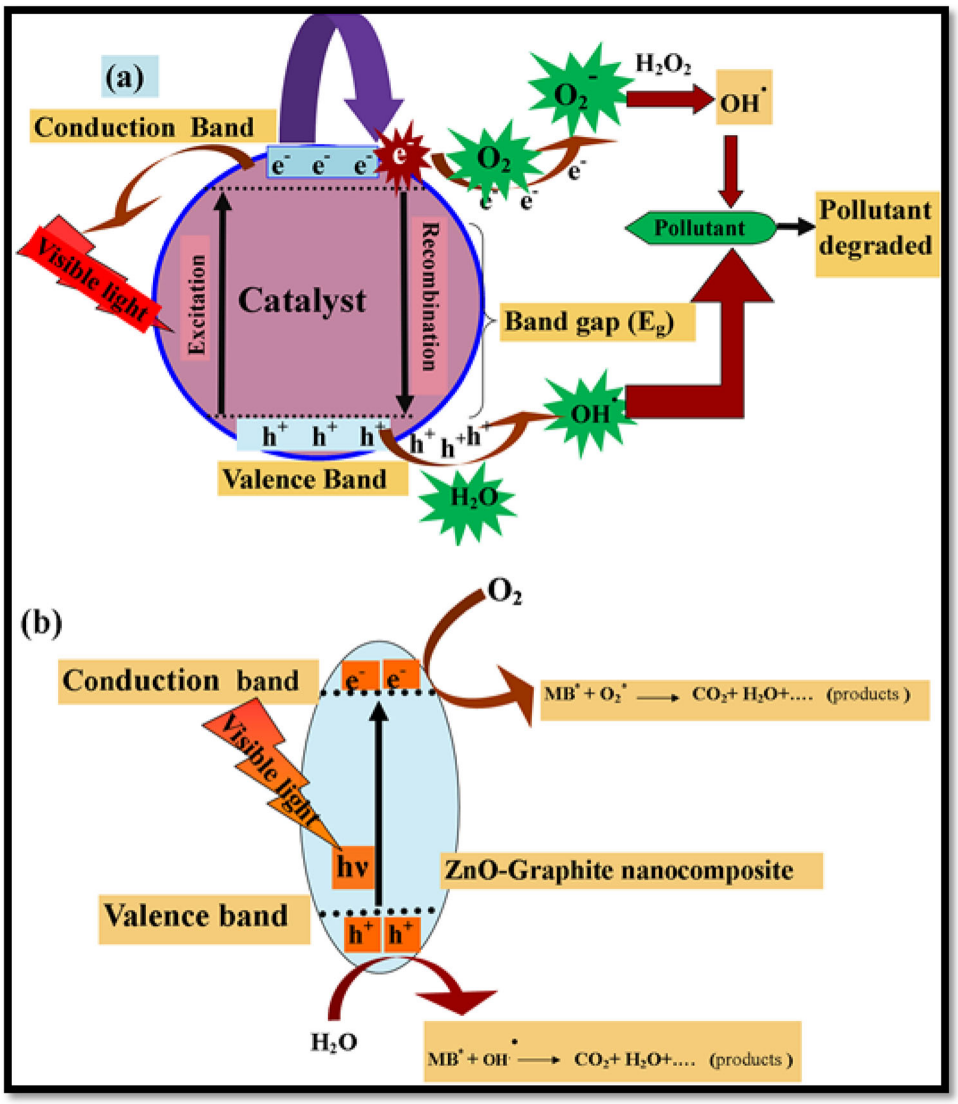


Fig. 7 **a** and **b** Mechanism of Photocatalysis degradation of MB dye

composite begins with oxidative and reductive processes, as described by equations [15, 31–36]. The proposed mechanism for the photocatalytic degradation of MB dye is explained as follows.

$$\mathrm{ZnO-graphite(NPs)} + h\upsilon(\mathrm{visible}) \rightarrow \mathrm{ZnO-graphite(NPs)} + (h^{+}\mathrm{VB} + \mathrm{e}^{-}_{\mathrm{CB}}) \tag{9}$$

$$h^{+}_{\mathrm{VB}} + \mathrm{H_2O} \rightarrow \mathrm{H^{+}} + \mathrm{OH^{0}} \tag{10}$$

$$2h^{+}_{\mathrm{VB}} + 2\mathrm{H_2O} \rightarrow 2\mathrm{H^{+}} + \mathrm{H_2O_2} \tag{11}$$

$$\mathrm{H_2O_2} \rightarrow 2\mathrm{OH^{0}} \tag{12}$$

$$2e^{-}_{\mathrm{CB}} + \mathrm{O_2} \rightarrow^{\mathbf{o}} \mathrm{O}^{-}_{2} \tag{13}$$

$$^{\mathbf{o}}\mathrm{O}^{-}_{2} + 2\,\mathrm{H^{+}} \rightarrow \mathrm{H_2O_2} \tag{14}$$

$$\mathrm{H_2O_2} \rightarrow 2\mathrm{OH^{\mathbf{o}}} \tag{15}$$

4 Conclusion

In this study, a ZnO/graphite nanocomposite was successfully synthesized via a combined sol–gel and physical grinding approach. The structural and morphological characteristics of the composite were confirmed using XRD and SEM, validating the effective

integration of ZnO with graphite. The electrochemical evaluation demonstrated that the ZnO/graphite composite delivered a maximum specific capacitance of 136 F g^{-1} at a scan rate of 10 mVs^{-1}. At a current density of 2 Ag^{-1}, the composite achieved an energy density of 20.8 $Whkg^{-1}$ with a corresponding power density of 797.64 Wkg^{-1}. The electrochemical behavior analysis indicated a dominant pseudocapacitive contribution, suggesting a synergistic interaction between ZnO and graphite that enhances charge storage capability. In addition to energy storage performance, the photocatalytic activity of the ZnO/graphite nanocomposite was investigated against the methylene blue (MB) dye. The composite exhibited efficient photocatalytic degradation, removing 86.2% of the MB absorption peaks under the tested conditions. Overall, the ZnO/graphite nanocomposite demonstrates high electrochemical performance, effective photocatalytic activity, cost-effectiveness, and environmental compatibility, making it a promising candidate for next-generation supercapacitor electrodes and photocatalytic applications.

Data availability The datasets analyzed during the current investigation are accessible upon reasonable request from the corresponding author.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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